

Tiling the Simplex: An Exact Quaternion Reading of Compositional Dynamics and Reproducible High-Dimensional Reconstruction

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Abstract

We present a deterministic, quaternion-chart construction for reading compositional trajectories. For a four-part composition, the three isometric-log-ratio (ILR) coordinates are identified with the imaginary part of a quaternion, so that a unit quaternion $q \in S^3 \cong \text{SU}(2)$ acts on this vector by the norm-preserving sandwich $v' = q v q^*$ —an exact $SO(3)$ rotation of the ILR space, reproduced to the IEEE floor (residual $\approx 4.4 \times 10^{-16}$). For higher dimension, a composition is tiled into overlapping exact four-part charts that share pivot components and reconstruct the full centered-log-ratio state through a connected atlas; structuring the atlas as a balanced tree bounds the graph diameter to $O(\log D)$ and, with it, the accumulated floating-point drift—giving measured reconstruction to $D = 10^6$ at $\approx 4.1 \times 10^{-12}$ floating-point residual (numerical reconstruction, not bit-exact identity). The construction is deterministic and hash-receipted under a stated canonicalization profile; the $D = 4$ conformance receipt is reproduced across implementations, while full cross-platform engine hash parity is treated in the companion tool paper (P3). We provide an open reference implementation, language-agnostic pseudocode, and a replication kit. This is a deterministic instrument whose exactness at four parts and reproducible high-dimensional reconstruction are demonstrated; the work makes no priority claim beyond the public timestamp.

1 Introduction

Compositional data—vectors of nonnegative parts carrying only relative information—live on the simplex with the Aitchison geometry [Aitchison, 1986, Egozcue et al., 2003]. We give an exact algebraic reading of compositional change at four parts and a deterministic, reproducible scheme to carry that exactness to high dimension by tiling. The construction is a descriptive instrument: it adds no model and asserts no dynamics; it reads what a trajectory does, exactly and reproducibly.

Contributions. (i) An exact identification of a four-part composition’s ILR coordinates with the imaginary part of a quaternion, on which a unit quaternion acts by the rotation $q v q^*$ to the IEEE floor; (ii) a connected overlapping-chart atlas that reconstructs the full clr state, with a balanced-tree variant whose $O(\log D)$ diameter bounds floating-point drift; (iii) a deterministic, hash-receipted reference implementation with a cross-platform conformance contract and a replication kit. We position the work as recognition and synthesis on established prior art (§2); we make no priority claim beyond the public timestamp.

*Human authorship for all claims; AI-assisted per HUF-STD-001. Correspondence and reproducibility kit: the `higgins-decomposition` repository.

2 Background and prior art

The log-ratio framework, the Aitchison inner product, and ILR coordinates are standard [Aitchison, 1986, Egozcue et al., 2003, Pawlowsky-Glahn et al., 2015]. Reconstruction of a composition from a connected log-ratio graph is established [Greenacre, 2019]; the graph-Laplacian recovery of a global signal from relative differences, exact iff the graph is connected with conditioning set by its diameter, is the synchronization / algebraic-connectivity picture [Singer, 2011, Fiedler, 1973]. Tree balances appear in PhILR [Silverman et al., 2017]. The novel elements are narrow and stated as such: the quaternion-exact $\text{ILR} \rightarrow S^3 \cong \text{SU}(2)$ reading of a *composition* (the nearest neighbours use a different sphere or a flat group; Scealy and Welsh, 2011, Birtea et al., 2022), and the deterministic locally-exact-plus-reproducible synthesis as an instrument. Greenacre et al. [2023] develops pairwise log-ratio selection on connected graphs for interpretability and variable reduction; the present construction is orthogonal—it retains the full set of within-chart log-ratios and recovers the global clr state exactly (in exact arithmetic), or to bounded floating-point drift (in floating-point), via the quaternion identification.

3 Method

3.1 The quaternion reading at $D = 4$

For a four-part composition $x \in \Delta^3$, the Helmert-ILR map sends x to $z \in \mathbb{R}^3$, which we identify with the imaginary quaternion $v = z_1 \mathbf{i} + z_2 \mathbf{j} + z_3 \mathbf{k}$ (a pure imaginary quaternion in $\text{Im}(\mathbb{H}) \cong \mathbb{R}^3$). A unit quaternion $q \in S^3 \cong \text{SU}(2)$ acts on this ILR vector by the sandwich

$$v' = q v q^*, \quad (1)$$

an exact $SO(3)$ isometric rotation of the ILR space, realised through the double cover $S^3 \cong \text{SU}(2)$. This is an Aitchison-isometric rotation of the ILR representation, *not* the Aitchison perturbation operation itself (which is a translation in ILR space). Measured against the direct $SO(3)$ rotation, the residual is $\approx 4.4 \times 10^{-16}$ (twice the double-precision machine epsilon), confirmed bit-identically on two unrelated reference inputs. Where exact-ILR coordinates are often considered hard to interpret—they are ratios of geometric means—the quaternion identification gives them a concrete geometric meaning: the unit quaternion $q \in S^3 \cong \text{SU}(2)$ acts on the ILR vector $v \in \text{Im}(\mathbb{H}) \cong \mathbb{R}^3$ by conjugation $q v q^*$, so the direction and magnitude of compositional change read off as a rotation axis and angle. The generators of the action satisfy the $\mathfrak{su}(2)$ relations $[G_i, G_j] = 2 \varepsilon_{ijk} G_k$ (Appendix B).

3.2 The atlas

A composition of dimension D is covered by overlapping four-part charts that share pivot components. Each chart is exact (§3.1). The charts form a graph whose nodes are parts and whose edges are the within-chart log-ratios.

3.3 Reconstruction and the connectivity condition

The full clr state is recovered by solving a sparse Laplacian system $A^\top A c = A^\top b$ built from the chart log-ratios; the solution is exact in exact arithmetic *iff* the chart graph is connected [Greenacre, 2019, Singer, 2011]. A disjoint atlas fails (measured reconstruction error $O(1)$); overlap is necessary.

3.4 Conditioning and the hierarchical atlas

The reconstruction’s floating-point error scales with the chart-graph diameter. A path atlas has diameter $O(D)$; a *balanced-tree* atlas has diameter $O(\log D)$, bounding the accumulated drift. This is the difference between an unusable high- D reconstruction and a reproducible one (Table 1).

3.5 Determinism and provenance

Every reading is computed by a fixed engine and stamped with a SHA-256 content hash over a canonical payload (round floats to 12 decimals; sorted keys; Python default separators; UTF-8). Identical inputs yield identical outputs and a matching hash. The engine-output hash uses this 12-decimal payload, whereas the $D = 4$ HS-EPS-1 conformance test uses a separate full-precision receipt over the exactness-residual fields, because those residuals ($\sim 10^{-16}$) live below the 12-decimal round; the conformance vector and engine-reference hash are pinned in the reproducibility kit. The cross-platform (Python/R) parity contract is the subject of the companion tool paper (P3).

4 Results (measured; reproducible)

At $D = 4$, qvq^* reproduces the Aitchison rotation to $\approx 4.4 \times 10^{-16}$ with norm preserved at the floor. The reverse tiling reconstructs the clr state with the residuals of Table 1; the per-chart $D = 4$ map is exact at the IEEE floor, while the high- D reconstruction is floating-point accumulation that grows with D —*not* a mathematical identity at scale. A balanced tree-atlas carries the reconstruction to $D = 10^6$ at $\approx 4.1 \times 10^{-12}$.

D (parts)	S^3 charts	reconstruction residual
16	13	1.3×10^{-15}
64	61	6.7×10^{-14}
256	253	3.0×10^{-13}
1024	1021	5.9×10^{-12}
10^6 (tree)	$\sim D/3$	$\approx 4.1 \times 10^{-12}$

Table 1: Measured reverse-tiling reconstruction residuals (floating-point accumulation, not bit-exact identity). Rows $D=16$ –1024 use the path atlas; the $D=10^6$ row uses the balanced-tree atlas. Reproducible from the replication kit; deterministic to a content hash.

Cross-platform conformance. We validated the $D = 4$ exactness with HS-EPS-1, a fixed-vector IEEE-754 conformance test: a single composition and unit quaternion, no random inputs, so every conformant `float64` implementation must return the same numbers. The sandwich qvq^* and the corresponding $SO(3)$ rotation agree to 1.11×10^{-16} (one unit in the last place); over 2000 random perturbations the worst-case residual is 4.4×10^{-16} . A content receipt over the exactness fields (SHA-256 06ccdb25...) reproduced *bit-for-bit across four independent implementations* (Appendix A), establishing platform-independent $D = 4$ exactness at the double-precision floor. Norm preservation holds to within one machine epsilon; its exact bit value depends on the norm-reduction order across BLAS/library versions and is therefore reported but excluded from the receipt.

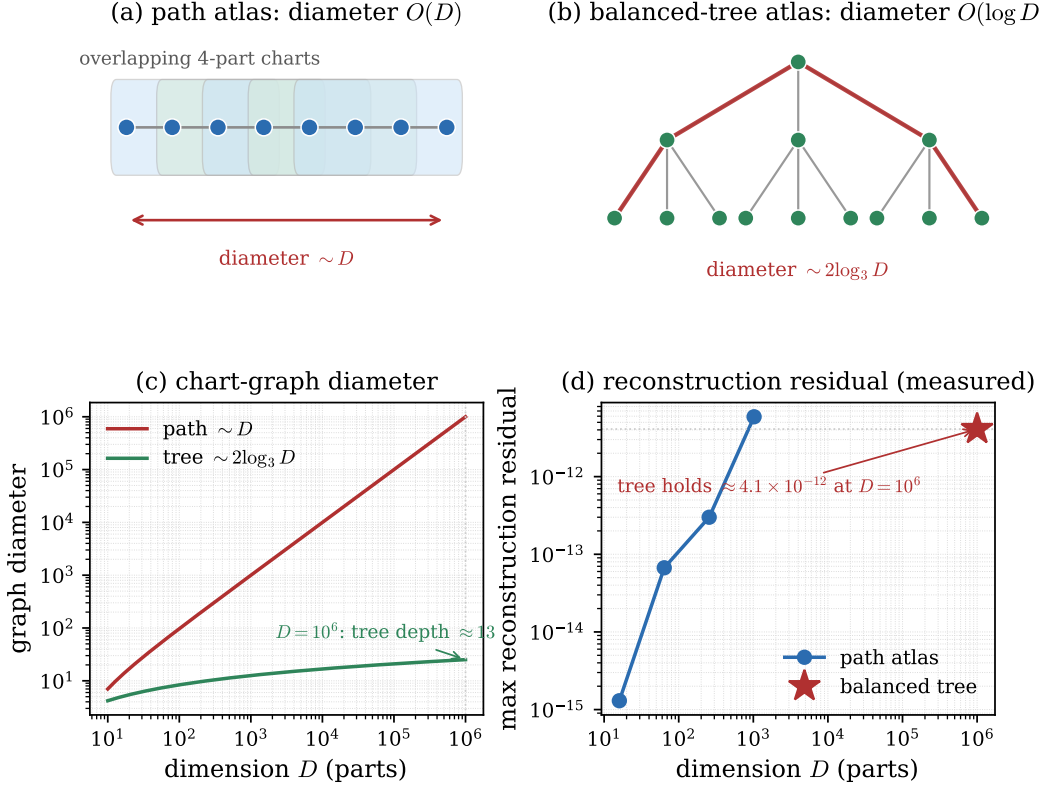


Figure 1: Chart-graph atlas and its conditioning. (a) A *path* atlas of overlapping four-part charts has diameter $\sim D$; (b) a *balanced-tree* atlas has diameter $\sim 2\log_3 D$. (c) Graph diameter vs. D : the path grows linearly while the tree grows logarithmically (tree depth ≈ 13 at $D = 10^6$). (d) Measured reconstruction residual (Table 1): the balanced tree holds the residual at $\approx 4.1 \times 10^{-16} - 10^{-12}$ even at $D = 10^6$, because the $O(\log D)$ diameter bounds the accumulated floating-point drift.

5 Discussion

The exact $D = 4$ reading and the reproducible high- D reconstruction together give a deterministic instrument for compositional trajectories that is locally exact and globally reproducible. Relation to existing compositional methods, and the kinematic and character-space readings built on this base, are developed in the companion papers (P4, P5).

6 Claim tiers

Tier 1 (verified): the $D = 4$ quaternion exactness ($\approx 4.4 \times 10^{-16}$); the measured reconstruction residuals (Table 1); intra-platform determinism + content hash. **Tier 2 (sound)**: the tiling/reconstruction synthesis as an instrument. **Tier 3 (to confirm)**: absolute terminological novelty (pending a final Scholar/ADS/patent pass); cross-platform byte-identical hash parity (the companion P3 reproduction). No claim of bit-exact high-dimensional identity; no priority claim beyond the public timestamp.

7 Reproducibility

Open reference implementation, language-agnostic pseudocode, an R port, and a replication notebook ship with the repository. Conformance: reproduce the canonicalization test-vector hash and the engine-reference hash on the reference matrix. Per-field numerical agreement at the IEEE floor is the substantive contract and is separable from byte-hash parity.

A Cross-platform conformance (HS-EPS-1)

HS-EPS-1 fixes one four-part composition $x = (0.4, 0.3, 0.2, 0.1)$ and one unit quaternion $q \propto (1, 2, 3, 4)$; with no random inputs, every conformant `float64` implementation must return the same core conformance receipt (norm preservation may differ by one ULP, the norm-reduction order being library-dependent). Table 2 reports five independent implementations. The sandwich-vs- $SO(3)$ residual is 1.11×10^{-16} in every case—one unit in the last place at the magnitude of the rotated components—and a content receipt over {machine ε , both residuals, verdict} (06ccdb25...) matches for all five. The only field that varies is norm preservation: it is either 0 or $\varepsilon = 2.22 \times 10^{-16}$ (one ULP at the magnitude of the unit norm), depending on the order of the square-root reduction in the host library. Both outcomes are within one ULP; the exact bit value is a property of the platform’s norm evaluation, not of the construction, so it is excluded from the conformance receipt.

implementation	sandwich residual	norm preservation	core receipt
1	1.11×10^{-16}	0	match
2	1.11×10^{-16}	0	match
3	1.11×10^{-16}	ε	match
4	1.11×10^{-16}	ε	match
5	1.11×10^{-16}	ε	match

Table 2: HS-EPS-1 across five independent `float64` implementations. The exactness residual and the core receipt are bit-identical; norm preservation differs by at most one ULP (norm-reduction order).

B The $\mathfrak{su}(2)$ generators

The 4×4 generators of the quaternion action are skew-symmetric with $\exp(\theta G)$ orthogonal of determinant +1, and close as $\mathfrak{su}(2)$ with structure constants $[G_i, G_j] = 2\varepsilon_{ijk}G_k$ (verified numerically; the explicit 4×4 generator matrices and the symbolic commutator check ship in the replication kit). The sandwich qvq^* is the adjoint action induced from the fundamental representation; the $D = 4$ case is the exact, associative division-algebra rung (S^3 is the unit-sphere group), which is why the reading is exact here and why higher dimension is reached by tiling four-part charts rather than a native high-dimensional rotor.

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